ILLINOIS TECH | College of Computing

ITMM 485 / 585
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Legal and Ethical Issues in Information Technology

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Ch8: Intellectual Property Issues
P1: Intro to IP, Software IP issues

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Announcements

- Assignment III
- Assignment IV
- Final Exam
- Content on Canvas

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Learning Objectives:

Upon completion of this lesson the students should be able to:

- Explain and discuss the philosophical and legal arguments used to protect intellectual property
- Describe the differences and similarities between the Free Software Foundation & the Open Source Initiative
- Explain how Creative Commons licensing of IP works
- Discuss the balance between public good and profit for IP rights holders

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Module Objectives:

Upon completion of this lesson the students should be able to:

- Define intellectual property (IP)
- Discuss protection of software as intellectual property through use of both patents and copyright

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Intellectual Property in Cyberspace

- What, exactly, is Intellectual Property (IP)?
- How have IP laws been challenged by cybertechnology and digital information?
- Before examining questions about IP, we can ask why property laws (in general) are important.

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Why Property Laws are Important

- Social scientists suggest that property Laws play a key role in (a) shaping a society and (b) preserving its order by establishing relationships between:
- individuals,
- different sorts of objects,
- the state.

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What Is (Tangible) Property?

- When discussing property, we tend to think of *tangible* items.
- Originally, "property" referred to land.
- Property now also includes objects that one can own, such as an automobile, articles of clothing, a laptop, a mobile phone.

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Property as a "Relational" Concept

- Property should not be viewed simply in terms of items or things (tangible or otherwise).
- Philosophers and legal theorists point out that property can best be understood as a *relationship between individuals in reference to things*, where three elements need to be considered:

1) an individual (*X*),
2) A "thing" or object (*Y*),
3) *X's* relation to other individuals (*A*, *B*, *C*, etc.) in reference to *Y*.

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Property as a Form of "Control"

- *X* (as the owner of property *Y*) can control *Y* relative to persons *A*, *B*, *C*, and so forth.
- If Harry owns a certain object (e.g. a Dell laptop computer), then Harry can control who has access to that object and how it is used.
- For example, Harry has the right to exclude Sally from using that laptop; or he could grant her unlimited access to it.
- Ownership claims involving "intellectual objects" (involving IP) are both similar to and different from ownership of tangible objects.

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Intellectual Objects

- The expression *intellectual object* can refer to various forms or instances of intellectual property.
- Intellectual property consists of "objects" that are not tangible.
- Intellectual objects represent creative works and inventions, i.e., the manifestations or expressions of ideas.

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Intellectual vs. Tangible Objects

- Tangible objects are *exclusionary* in nature.
- If Harry owns a laptop computer (a physical object), then Sally cannot, and *vice versa*.
- Intellectual objects, such as software programs, are *non-exclusionary*.
- If Sally makes a copy of a word-processing program (that resides in Harry's computer), then both Sally and Harry can possess copies of the same word-processing program.

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Intellectual vs. Tangible Objects

- The sense of *scarcity* that applies to tangible objects, which often causes competition and rivalry, need not exist for intellectual objects.
- For example, there are practical limitations to the number of physical objects that one can own.
- There are also limitations (natural and political) to the amount of land that can be owned.
- Intellectual objects can be easily reproduced.
- Countless copies of a software program can be produced – each at a relatively low cost.

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Ownership of Intellectual vs. Tangible Objects

- Legally, one cannot own an idea in the same sense that one can own a physical object.
- Governments do not grant ownership rights to individuals for ideas per se.
- Legal protection is given only to the tangible *expression* of an idea that is creative or original.

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Ideas vs. Expressions of Ideas

- If an idea is literary or artistic in nature, it must be expressed (or "fixed") in some tangible medium in order to be protected.
- A “tangible medium” could be a physical book or a sheet of paper containing a musical score.
- If the idea is functional in nature, such as an invention, it must be expressed in terms of a machine or a process.
- Authors are granted copyright protections for expressions of their literary ideas, while inventors are given patent protection for their inventions.

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Why Protect Intellectual Property?

- One answer is: Our current laws say that intellectual property should be protected.
- But we can ask: On what philosophical grounds are our property laws themselves based?
- In Anglo-American law, philosophical justification for intellectual property rights is grounded in two different types of views:

1) natural rights,
2) conventional (or constructed) rights.

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Protecting Intellectual Property

- One theory holds that a property right is a "natural right," to which individuals are justified for the products that result from their labor, including intellectual objects.
- The other theory views property rights as a social construct designed to encourage creators and inventors to bring forth their artistic works and inventions into the marketplace.

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Software as Intellectual Property

- Should computer programs be eligible for patent protection?
- Should they be protected by copyright law?
- Do they deserve both, or perhaps neither, kind of protection?
- Computer software consist of lines of programming code (or codified thought).
- It is not expressed or "fixed" in a tangible medium in a way that literary works are.

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Software as Intellectual

- A program's *source code* consists of symbols.
- Its *object code* is made up of "executable images" that run on the computer's hardware after they have been converted from the original source code.
- Initially, it was not clear that software programs should be given copyright protection.
- Some argued that computer programs are more like inventions that can be patented.

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Software as Intellectual Property

- Software programs also resemble algorithms, which, like mathematical ideas or "mental steps," are not eligible for patent protection.
- Initially, computer programs were not eligible for either copyright or patent protection.
- Eventually, however, *both* copyright and patent protections were granted to software programs.

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The Case for Protecting Software as a form of Intellectual Property

- The software industry has made the following kind of argument for why software should be protected with intellectual property rights.
- **PREMISE 1.** Stealing a tangible object is morally wrong.
- **PREMISE 2.** Making an unauthorized copy of proprietary software is identical to stealing a tangible object.
- **CONCLUSION.** Making unauthorized copies of proprietary software is morally wrong.

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Protecting Software (Continued)

- If we apply the rules for logical validity (in Chapter 3), we see that this argument is *valid* because of its logical form.
- Consider that *if* both Premises 1 and 2 are assumed true, the conclusion cannot be false.
- Even though the argument's form is valid, we could still show the argument to be *unsound* if either or both of the premises are false.

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Protecting Software Continued)

- Premise 1 (in the above argument) is fairly straightforward, and few would question its truth.
- But Premise 2 is more controversial and thus we can question whether it is empirically true.
- For example, is duplicating a software program *identical* to stealing a physical item?
- Consider that software programs, like other intellectual objects, are nonexclusionary; so my having a copy of Program X does not exclude your having a copy of that program, and vice versa.
- Because the truth of Premise 2 is questionable, we cannot infer that the above argument is sound.

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Protecting Software (Continued)

- But even if the original argument turns out to be unsound, it does not follow that its conclusion – "Making unauthorized copies of proprietary software is morally wrong" – is false.
- For example, the argument's conclusion could be true for reasons other than those stated in the premises.
- Consider that even if duplicating software is not identical to stealing physical property, it could still cause harm to the property owner because copying the proprietary software program (like the theft of someone's physical property) deprives the property owner of the legitimate use of his or her property.

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Protecting Software (Continued)

- One could also argue, for example, that unauthorized copying is harmful because it is a misuse, misappropriation, or “unfair taking” of another person’s property against the property owner’s will (Spinello, 2008) .
- So, there could be alternative reasons why the original argument’s conclusion can be true, despite the fact that its second premise may be false.

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Intellectual Property Protection Schemes

- We examines four schemes for protecting intellectual property:

1) *Copyrights*;
2) *Patents*;
3) *Trademarks*;
4) *Trade secrets*.

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